



Healthy Skepticism Day 3



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The content does not constitute medical advice and is not a substitute for professional diagnosis, treatment, or care. Nothing presented here should be used to make decisions about your health or the health of others.

Always seek the advice of a qualified physician or other licensed healthcare provider with any questions you may have regarding a medical condition, test result, or treatment. Never disregard professional medical advice or delay seeking it because of something discussed in this presentation.



The Five Faults

1

Relative reporting

Day 1

What does a 36% improvement from taking a drug really mean?

2

Probability reversal

Day 2

What does a positive COVID test really mean?

3

Arbitrary categories

Day 3

How concerned should you be if your systolic BP goes from 119 to 120?

4

The Bernoulli fallacy

Day 3

Why does the published medical advice keep changing?

5

The causal leap

Day 4

Does bird watching prevent dementia?

Once you see them, you can't unsee them.



Today's Two Faults

3

Arbitrary categories

Blood pressure, body weight, and where the lines fall

4

The Bernoulli fallacy

The 0.05 line, and what a study can and cannot tell you



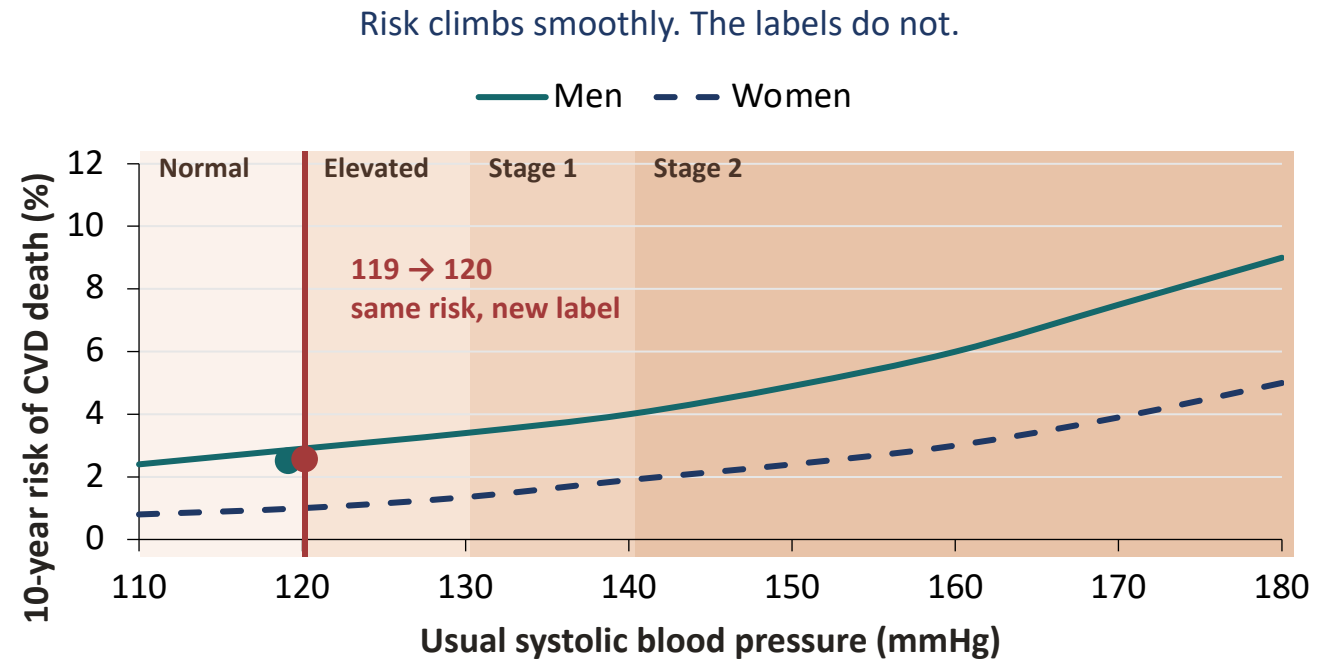
FAULT 3

Arbitrary Categories

Read the curve, not the cutoff

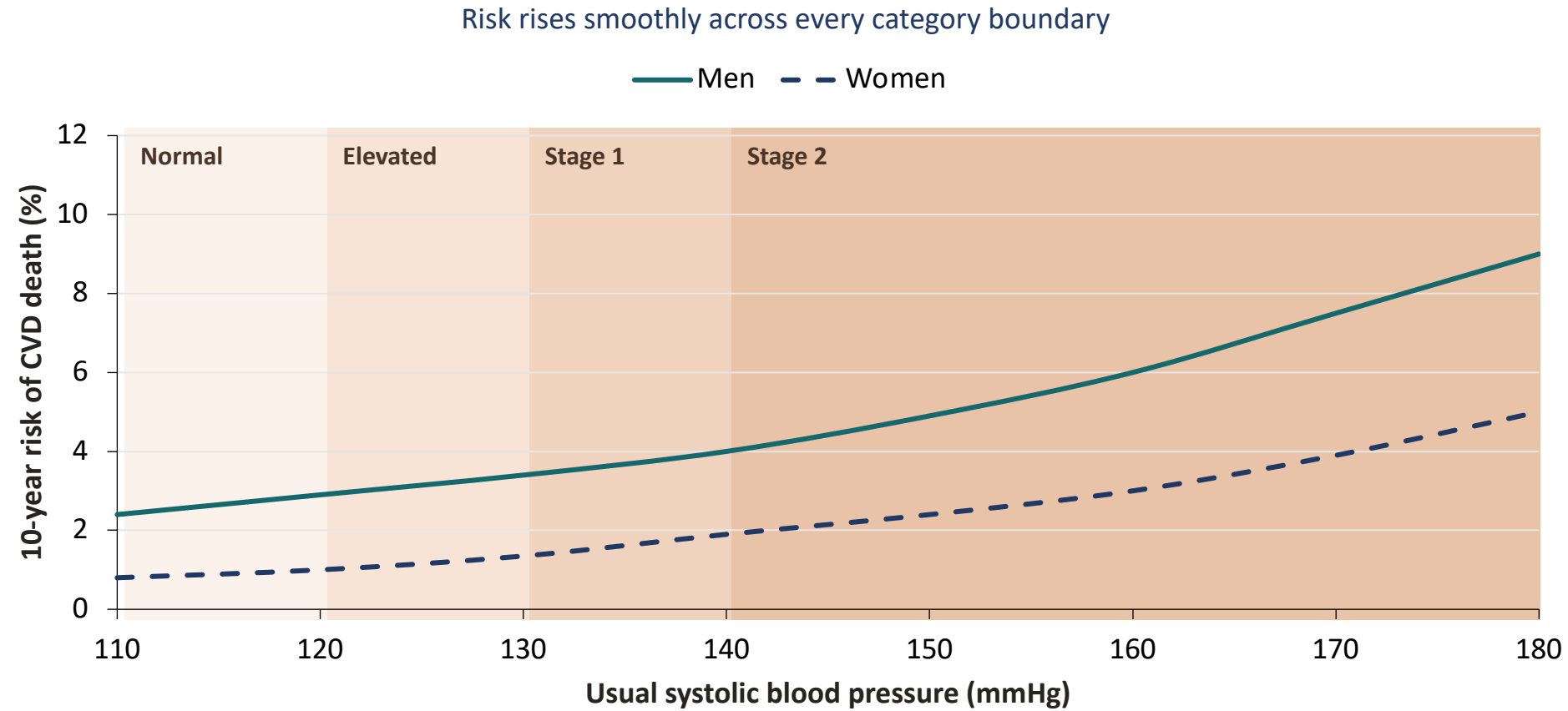
Where Does “High” Begin?

- Systolic this year: 120. Last year: 119.
- Nothing about you changed. The label did.



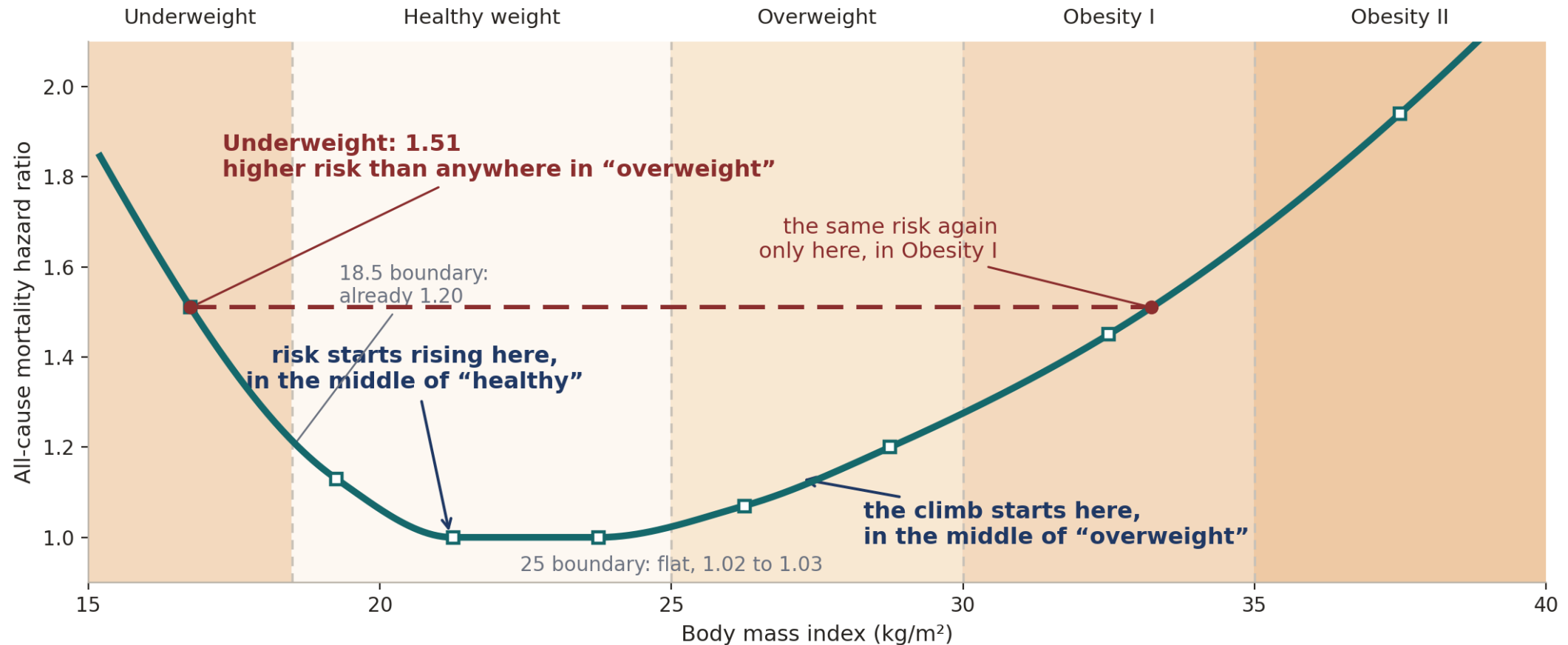
One mmHg moved you across a line. It did not move your risk.

Systolic Categories Don't Mean Much



SCORE risk charts, low-risk regions: Conroy RM et al., Eur Heart J 2003;24:987-1003. Category boundaries: ACC/AHA 2017 hypertension guideline.

BMI Categories Are Worse

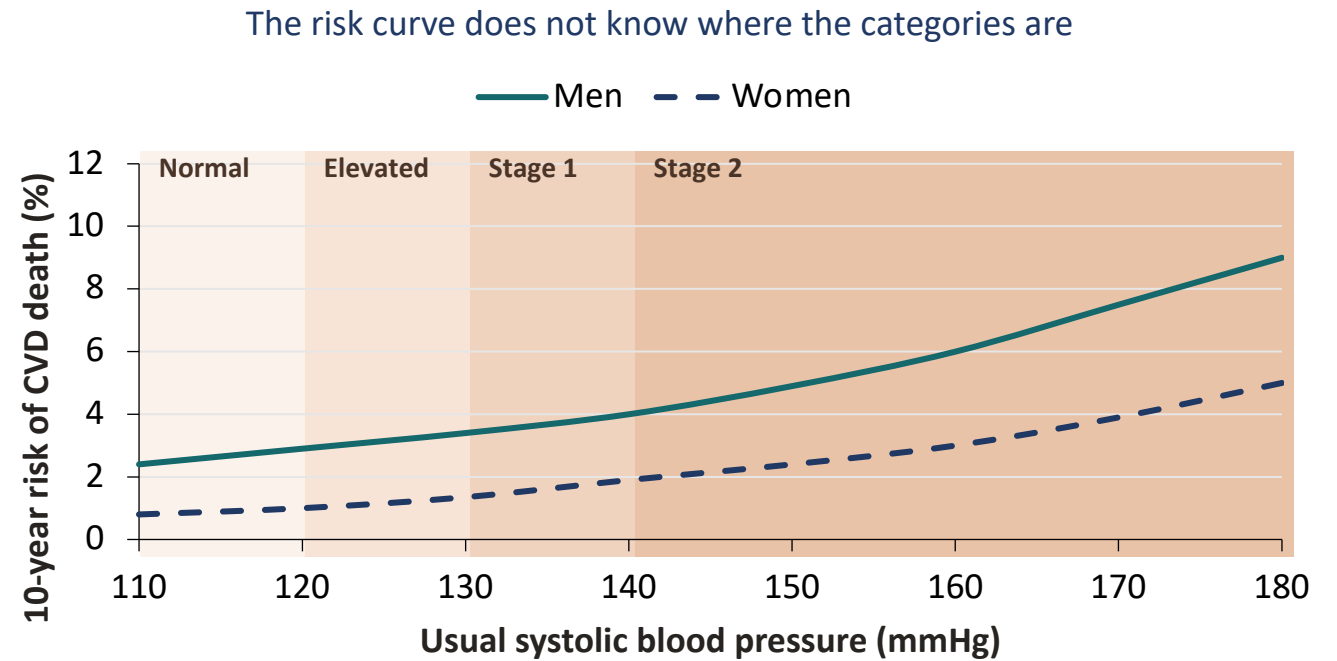


The risk transitions sit in the middle of the categories; the boundaries sit where nothing happens.

Global BMI Mortality Collaboration, Lancet 2016;388:776-786. All-cause mortality hazard ratios relative to BMI 22.5-25, never-smokers; band-midpoint values, shape-preserving interpolation.

So Why Have Categories at All?

- A box is a decision rule: triage in three seconds.
- The trouble: the box gets read as a fact about your body.
- A box cannot say how much risk you carry.



Ask where you sit on the curve, not which side of the line.



FAULT 4

The Bernoulli Fallacy

Why medical advice keeps changing

Published Papers with Findings That Could be Repeated

Only 6 of 53 landmark cancer findings were confirmed on rerun.

Just 35 of 97 psychology findings stayed significant on rerun.

This is the Irreproducibility Crisis

An Important Aside: $P(A | B)$ is The Core Element of Inference

- It is called 'conditional probability'
- It means the probability of A if B is true
- Sometimes shortened to the probability of A, given B
- Depending on the circumstances, B may be observed, believed, or imagined for argument's sake
- **Examples:**
 - Courtrooms: $P(\text{Criminal guilt} | \text{evidence})$
 - Gambling: $P(\text{Rolling 7 with 2 dice} | \text{Dice are fair})$
 - Science: $P(\text{Tested hypothesis} | \text{data})$
 - Causality: $P(\text{drowning} | \text{warm weather})$

For the course, you only need to be aware of the concept

Sometimes We Want to Reverse Conditional Probability

Day 2 had 2 examples

- As we saw in Day 2, we can reverse $P(A|B)$ to $P(B|A)$ if we also have $P(B)$
- $P(B)$ is sometimes called the 'prevalence' or 'prior' depending on the circumstances
- For example, in the mammogram example
 - $P(\text{called back} | \text{cancer}) = 87\%$
 - $P(\text{cancer}) = 0.6\%$ (6 in 1000)
 - We computed using ratios $P(\text{cancer} | \text{called back}) = 4.4\%$

This sort of calculation, reversing using the prior, is called 'Bayesian Analysis'

The Fallacy Stems From Frequentist Reasoning

- They want to know if a hypothesis is true.
- Rather than compute $P(\text{hypothesis} \mid \text{data})$, they compute $p = P(\text{data} \mid \text{hypothesis is false})$
- If p is low enough, they say the observations are 'statistically significant' and submit the result for publication.
- In medicine and the social sciences, the significance criterion is $p < .05$: if the hypothesis were false, the data would arise only 5% of the time.
 - In words: a false hypothesis would make this data very surprising, so treat it as true

This criterion means one would expect about 1 in 20 published experiments to be not reproducible.

The Drawbacks

- 1 **The significance setting is arbitrary, and often far too high.** There is no formula or theorem that defines 'significance' .
- 2 **Significance is not informative.** Knowing $P(\text{hypothesis} \mid \text{data})$ enables risk-based decision support, like gambling; the p-value does not.
- 3 **It invites p-hacking.** Searching subsets of the data for those that meet the criterion by luck alone.

Together these feed the replication crisis

Medical and Social Science Sets the p-value Far Too High for Significance

Medicine cannot wait for five-sigma data: patients are scarce, trials costly.

WHAT PHYSICS DEMANDS

$$p < 0.000000287$$

Five sigma: one chance in 3.5 million. The Higgs waited for that bar.

WHAT MEDICINE AND SOCIAL SCIENCE ACCEPT

$$p < 0.05$$

One chance in twenty. A no-effect test clears it 1 time in 20, by luck.

1 in 20 is the rate per test of nothing. The share of false findings is a reversal question: how many tested ideas were true? The prevalence, yet again.

So Why Do They Adopt Frequentism?

- **Sometimes there is no prior:**
 - Industrial quality control
 - Census survey sampling
 - Particle physics (which sets the criterion at $p < 0.000000287$)
 - Not using a prior is considered more objective and honest by its proponents.
- **As described in the book**, a more rigorous standard disrupts the academic publication world.

Evidence of p-hacking

Only 6 of 53 (11%) landmark cancer findings were confirmed on rerun.

Just 35 of 97 (36%) psychology findings stayed significant on rerun.

These are much lower than chance (95%)

Evidence for ‘p-hacking’, aka ‘data torturing’

The observed failure rate is far above the honest floor.

1 in 20

the failure rate honest testing would produce on its own, with no torturing at all.

47 of 53

landmark cancer findings that failed to repeat. Psychology: about 2 in 3.

The failure rate is too high for luck alone.

The Baby Aspirin Recommendation Change

Trials stopped early tend to stop on a lucky high. On the retests, p never crossed the line again.

1988

Physicians' Health Study: aspirin cut first heart attacks 44%, $p < 0.00001$. Stopped early for benefit. Decades of daily-aspirin advice followed.

2018

The retests, ARRIVE and ASPREE: no net benefit (ARRIVE $p = 0.60$), more major bleeding. USPSTF 2022: do not start after 60.

The advice changed because the finding would not repeat, and the harm outweighed the benefit

Autism: From a p-Value to Policy

31

autism cases, parent-reported, not
medical records

3x

the reported odds, in boys only, not
the full sample

$p < 0.05$

so it was written up as a finding

- 1 **One study:** 31 parent-reported cases.
- 2 **One number:** p below 0.05.
- 3 **One citation:** the CDC's Autism and Vaccines page.
- 4 **One committee:** ACIP, twice, days before a vote.
- 5 **One policy:** universal newborn hepatitis B vaccination, ended.

Never replicated. In May 2026 the journal retracted the paper, months after the vote.



What We've Learned

Two faults, one shape.

1

1 · Arbitrary categories

The trap: A line drawn across a continuous measurement is read as a fact about your body.

Ask: What is my risk with my measurement?

2

2 · The Bernoulli fallacy

The trap: p below 0.05 is read as proof the effect is real, and that has made the medical literature unreliable.

Ask: Have these studies been replicated?



Questions, and a conversation

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Backup Slides



A Line You Have Already Met

- Arbitrary categories came from clinical medicine: BMI 25, systolic 130.
- Statistics draws the same kind of line on itself. A study is "significant" if p falls below 0.05, and not otherwise.
- p is a continuous quantity. 0.05 is a convention, not a discovery. R. A. Fisher drew it in 1925 because it was convenient.
- Everything Fault 3 taught about cutoffs applies to the line the research itself is graded on.



Too Few Cases, Read as No Effect

- [Placeholder: ASCOT-LLA's women.]
- Only 36 heart events among women in the whole trial. The label calls the result inconclusive.
- Correspondents read that as evidence the drug does not help women.
- The investigators replied that 36 events cannot show much, and that absent evidence of a different effect, the whole-trial result is the best estimate for women too.
- Source: Sever PS, et al. Lancet 2003;361:1986.



Sixty Side Effects at Once

- [Placeholder: the 2026 statin label meta-analysis.]
- More than sixty labelled side effects tested, with a strict bar to avoid finding things by luck.
- Six survived. The conclusion drawn was that the labels should be trimmed.
- A strict bar also means real effects get missed, and that rate was never reported.
- Source: Cholesterol Treatment Trialists' Collaboration, Lancet 2026;407:689-703.